

Systems with built-in intelligence have made our life much easier, safer and trendy. So have the smart sensors that go into these systems. Vivek Sharma, regional vice president, Greater China and South Asia-India Operations, and director, India Design Centres, STMicroelectronics, in a conversation with Shweta Dhadiwal Baid of EFY Bureau, throws light on the smart sensors and their market in India

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SMART SENSORS ARE THE SOURCE OF CURIOSITY FOR MANY DESIGNERS

What is a smart sensor? How does it differ from a normal sensor?

Anything that is smart can make its own decision. Similarly, a smart sensor has built-in decision-making capability. The decision making capability is usually provided by the microcontroller that sends a control signal; for example, to stop a process or to relay the information to a central node for monitoring. Smart sensors are useful in a variety of areas ranging from industrial safety, security, hazard and accident prevention to early detection of unusual mechanical behaviours of machines.

Why are smart sensors important? What impact do they have on the real-life situations?

Smart sensors can be deployed in manned as well as unmanned situations. These have found applications in several real-life situations where response time is critical; for example, in fire detection. A temperature sensor based smart sensor connected with ZigBee can be used to detect forest fires and take preventive action sooner than otherwise.

Smart-sensor-based security systems use a simple PIR

sensor and transmit the data to a remote location, either to a computer on Wi-Fi or a GPRS mobile phone. A ZigBee-based home automation system could be connected to the Internet gateway either wirelessly or through wires in order to facilitate remote monitoring of the home, factory or any other installation.

In a smart-sensor-based building management system, the decision to sprinkle water in a particular area can be taken based on the temperature rise recorded in the vicinity.

MEMS-based smart sensors can collect information on aberrations in the elderly and transmit it to a doctor for a faster preventive or correction action.

Since the intelligence is provided by the microcontrollers, it is possible to have any kind of configuration for online or offline analysis of the recorded data.

What is the contribution of the semiconductor industry in this area?

The semiconductor industry has by far played the most important role in smart sensors. The computing element